BED

LECTURE

01

ngee ann school of infocomm technology

Introduction to Backend Development

Backend Development

Diploma in IT

Year 2 (2026/27), Semester 1

Topics

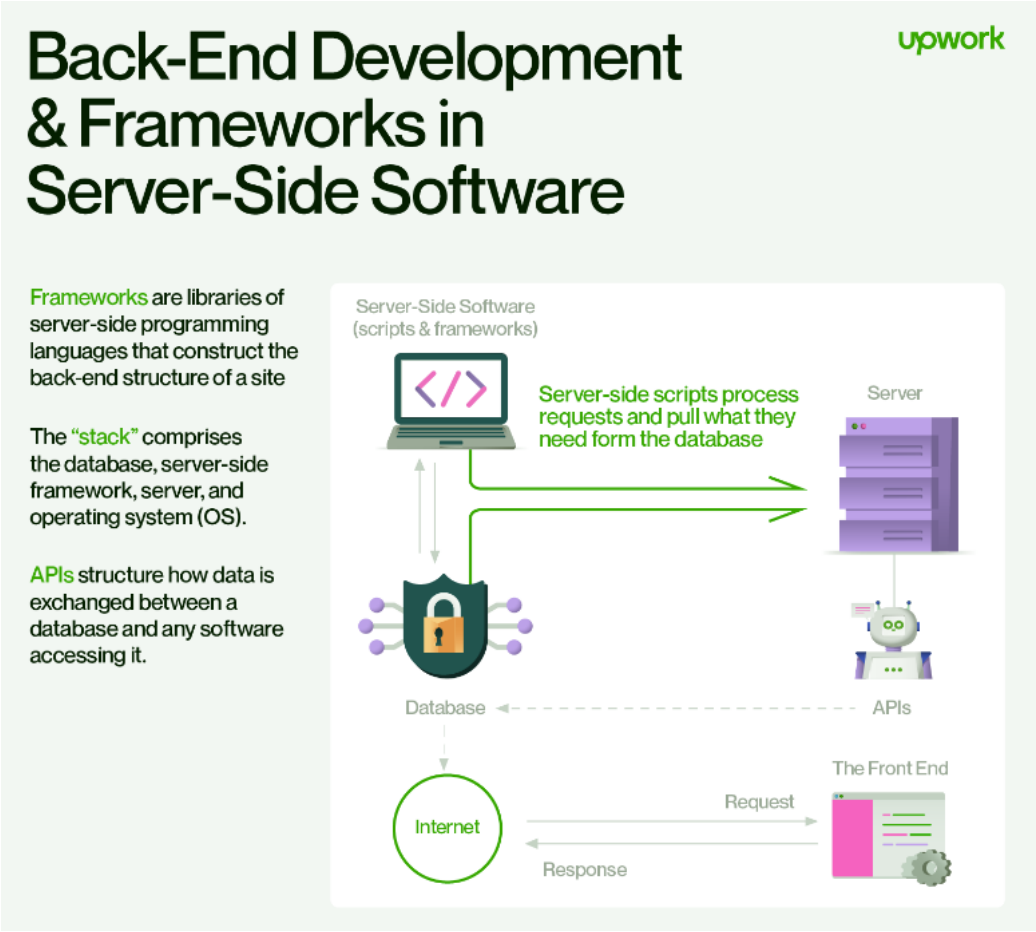
- What is a Backend Development
- Front-End vs Back-End Development Technologies
- Node.js and Express

Reading Reference

- **A beginner's guide to Backend Development**
  https://www.upwork.com/resources/beginners-guide-back-end-development
- **Frontend vs Backend**
  https://www.geeksforgeeks.org/frontend-vs-backend/
- **Explore Web Development with Node.js**
  https://www.linkedin.com/learning/paths/explore-web-development-with-node-js

What is a Backend Development?

- Working on the Server Side Software
- Focuses on everything you can't see on a website – databases, back-end logic, application programming interface (APIs), architecture and server



(Courtesy of Upwork Global Inc., extracted from https://www.upwork.com/resources/beginners-guide-back-end-development

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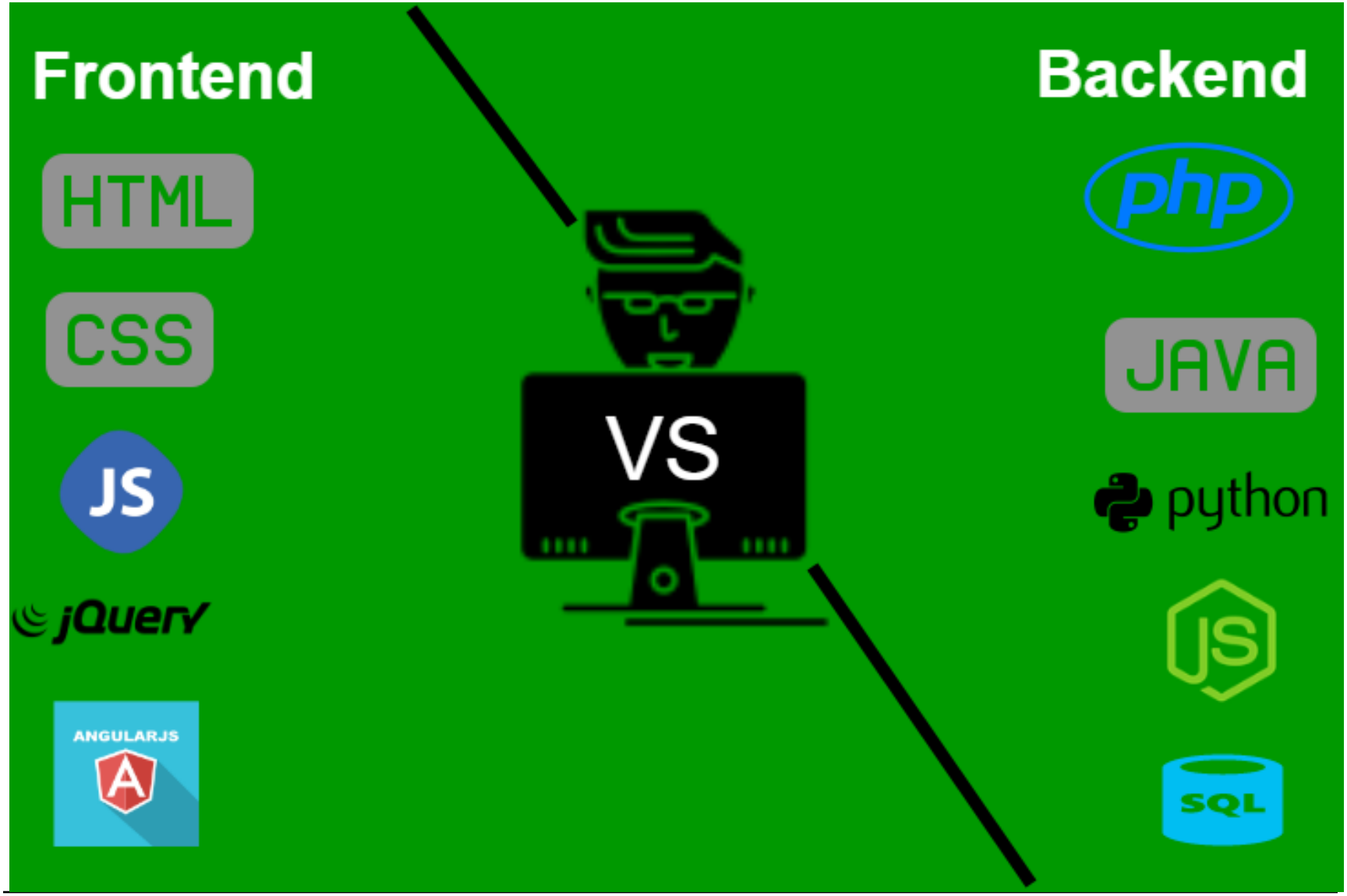
Front-End vs Back-End Development



Front-End vs Back-End Development

Comparison	Front-end Development	Back-end Development
Definition	Focuses on the visual parts of a website or web app.	Focuses on including building APIs and ensuring app performance and security.
Aim	Ensures web pages and apps are accessible and responsive on all devices.	Ensures servers, databases, and APIs function reliably, securely, and efficiently
Skill Set Required	Requires HTML, CSS, and JavaScript.	Requires languages like Python, Ruby, PHP, Java, etc.

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Front-End vs Back-End Development

Comparison	Front-end Development	Back-end Development
Additional Skills	Involves creativity and a good eye for design to build user-friendly interfaces.	Relies on logical thinking and problem-solving to ensure everything works smoothly.
Dependency	Often depends on the back-end for data and functionality, except in simple static websites.	Able to function independently, and changes made here typically don't impact the front-end directly.
Frameworks	React, Vue.js, Angular, Svelte, and Next.js	Node.js Express, Ruby on Rails, Java Spring Boot and Python FastAPI
Development Team	Focuses on designing and developing the user interface and experience of the application.	Supports the front-end by building and maintaining the server-side logic, APIs, and databases.

Sources: https://www.monocubed.com/blog/front-end-vs-back-end-web-development/

Common Back-End Technologies

- **Node.js** – Enables server-side development with JavaScript. Ideal for developing fast and scalable server-side applications, especially for APIs.
- **PHP** – One of the most widely used server-side scripting languages. Known for its simplicity and commonly used for content management systems (CMS) like WordPress.
- **Python** – Highly favored for applications involving data analytics, machine learning, and automation.

Common Back-End Technologies

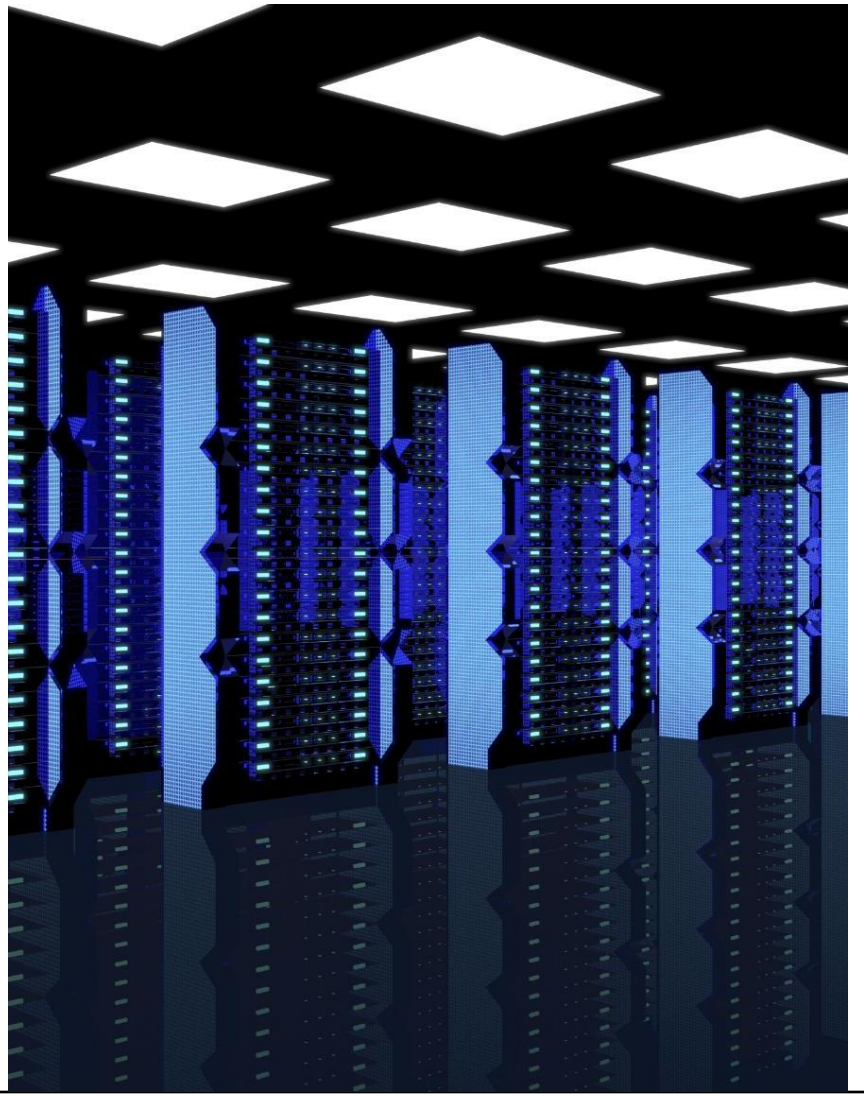
- **Ruby on Rails** – An open-source framework using Ruby, ideal for rapid web development, often used by startups for quick deployment.
- **ASP.NET** – Microsoft framework for building secure, scalable web applications, commonly used for corporate-level projects.

Why Choose Node.js and Express?

- Node.js allows JavaScript usage on both client and server sides.
  - This simplifies the learning process for learners.
- Streamlines backend development with its minimalist framework
  - Making it easier for learners to focus on core concepts.
  - Helping learners to grasp essential back-end concepts quickly.
- Popularity in the industry also ensures that students gain relevant skills.

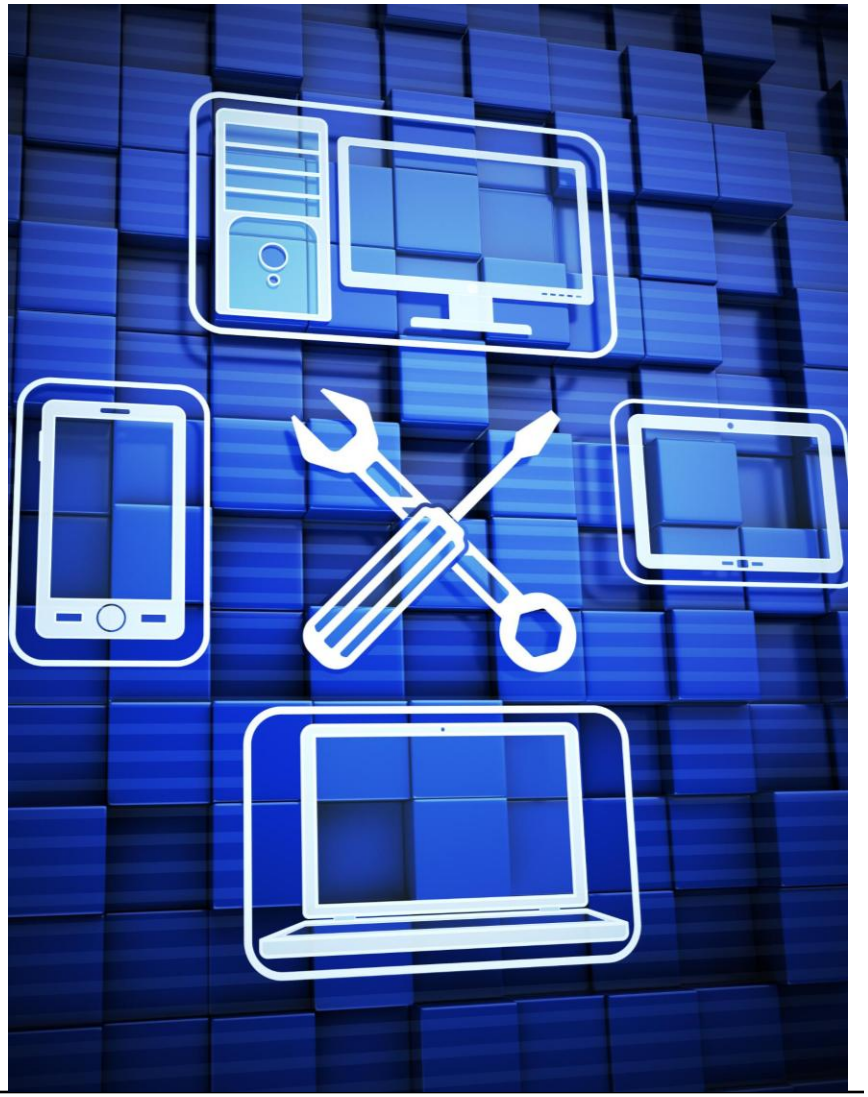
What is Node.js?

- Node.js is a server-side development tool that uses JavaScript.
- It's open-source, allowing global contributions for continuous improvement.
- Compatible across platforms like Linux, macOS, and Windows.
- Includes npm, a package manager with many libraries.



What is Express framework?

- Express.js is a lightweight and flexible web framework for Node.js, widely used for building web applications and APIs.
- Simplifies Development:
  - Offers a clear and easy way to quickly build backend apps
- Time-Efficient:
  - It's easy to learn and speeds up backend development.
- These benefits make Express a valuable tool for creating scalable backend applications.



Node Package Manager (npm)

- npm is essential for managing dependencies in Node.js projects.
- It simplifies packages installation and version control.
- Packages are reusable code modules that add features like routing, authentication and database access.

package.json with example

- The package.json is a configuration file that holds essential information about the project like metadata and dependencies.

```
{
  "name": "myapp",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^4.17.1"
  }
}
```

Summary

- Backend Development
  - Focus on server-side, database, backend logic and API development
- Frontend vs Backend Development
- Node.js
  - Open-source, cross-platform javascript server-side development tool
- Express
  - A popular web application framework for Node.js